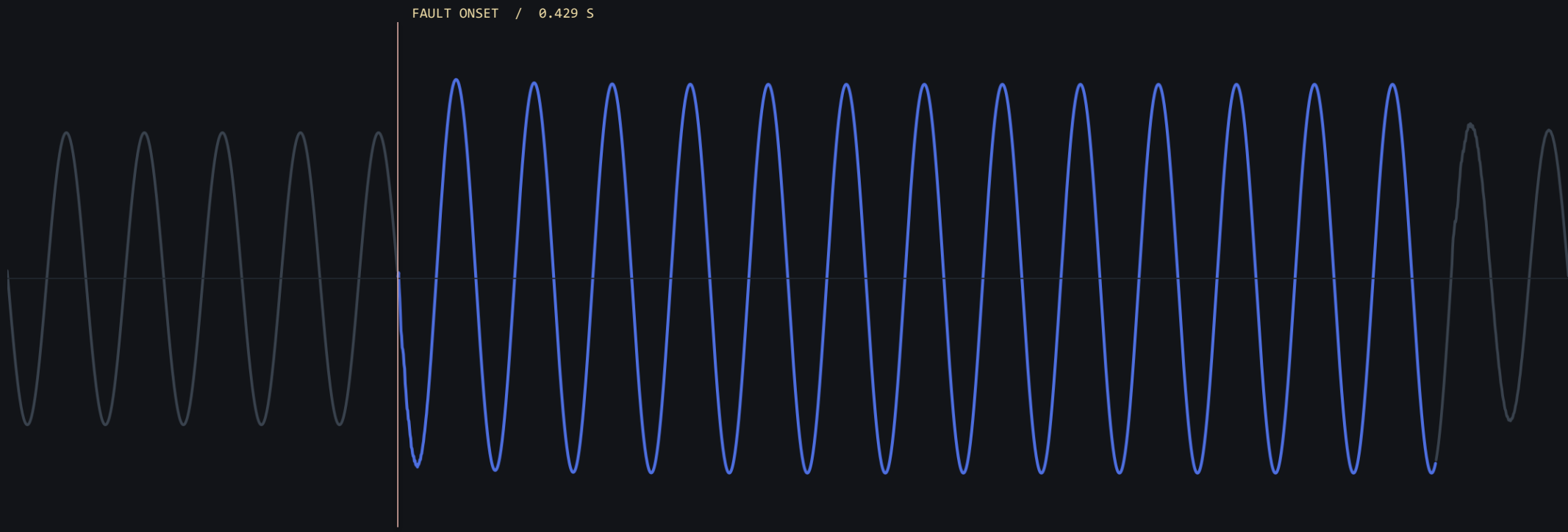
VIDYUT

INTERNAL TECHNICAL REVIEW

GROUP EE14 / THAPAR INSTITUTE OF ENGINEERING AND TECHNOLOGY

DOCUMENT 001
SHEET 01 / 10



SPECIMEN 0000 / PHASE L1 CURRENT / 0.33-0.73 S / +/- 1.1 KA

Fault classification from recorded transmission-line waveforms

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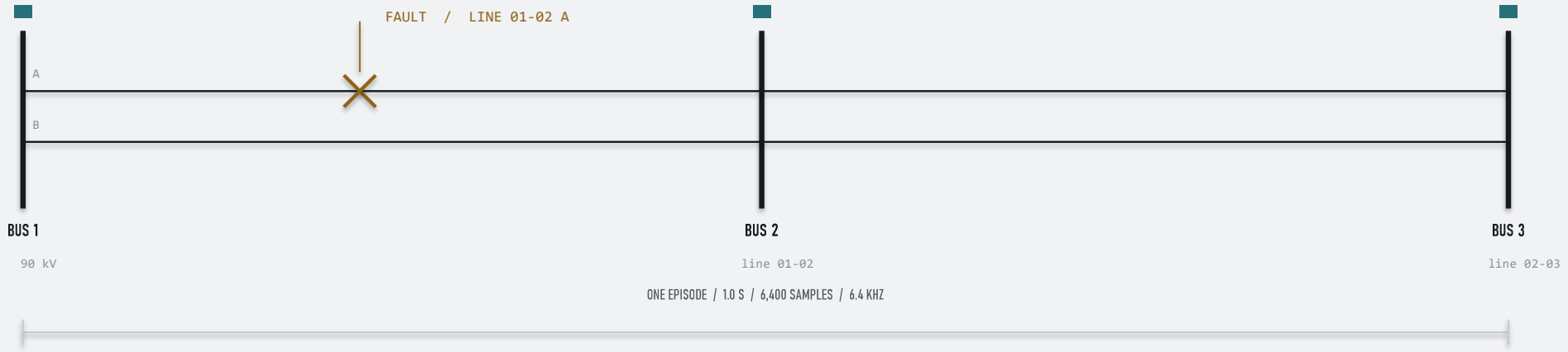
GROUP EE14
AI FOR ENGINEERS
DEPT. OF ELECTRICAL ENGINEERING
SESSION 2026-27

REVIEW: DR. KRISHNA KUMAR GUPTA

PROBLEM STATEMENT

A fault is a change in the shape of the current.

8 MEASUREMENT LOCATIONS / 6 CHANNELS EACH / 6.4 kHz



FOUR LABELS

LLL	three phases joined	sc_type 0
LL	two phases joined	sc_type 1
SLG	one phase to earth	sc_type 2
LLG	two phases to earth	sc_type 3

OBJECTIVE

Train classifiers that assign one of four fault categories to a 40 ms window of the 48 recorded channels, and evaluate them on episodes the models never saw.

The four labels differ only in which phases complete the path and whether earth is involved. Inside a 40 ms window that is a phase-current pattern, not a threshold.

NOT A RELAY / NO TRIP DECISION / NO FAULT DISTANCE

EXPERIMENTAL DATASET

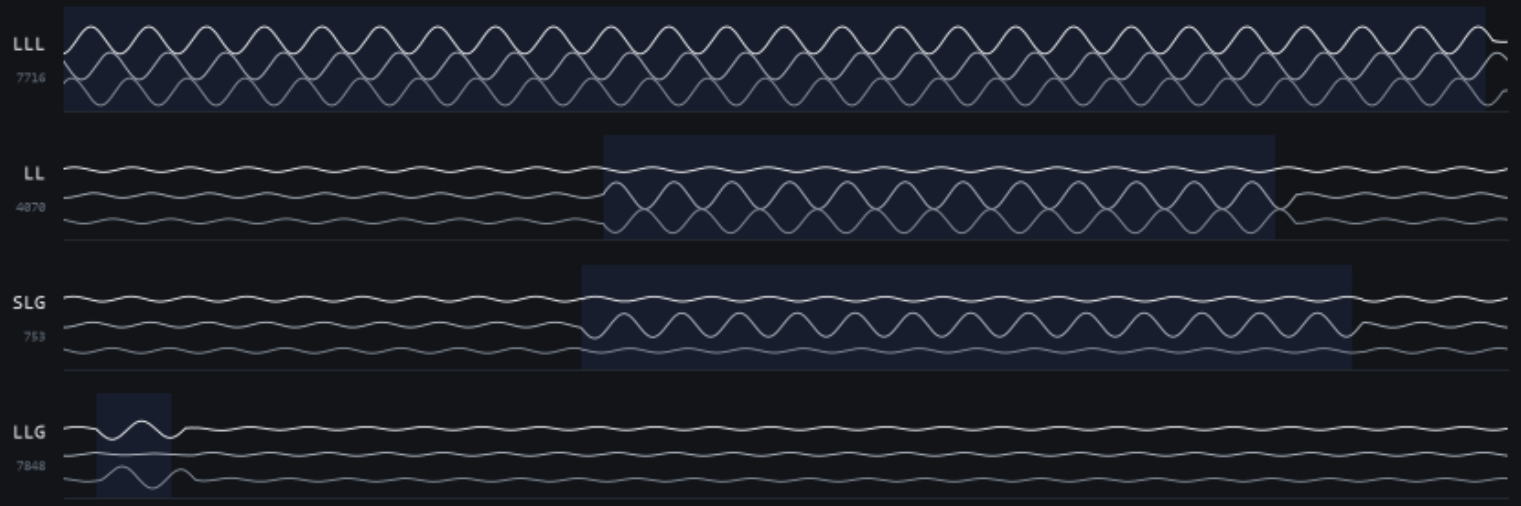
Nine thousand faults, four signatures.

PROTECT-90

9,022

EMT SIMULATION EPISODES

SYSTEM	90 kV, double circuit
CHANNELS	48, at 8 locations
SAMPLING	6.4 kHz, 128 per cycle
EPISODE	1.0 s, 6,400 samples
EVENTS	mean 226 ms, 828 short

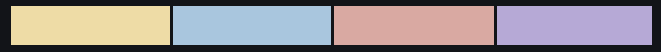


FOUR EPISODES / THREE PHASE CURRENTS EACH / BAND = FAULT INTERVAL / Z-SCORED

WHY THE SIGNATURES SEPARATE

A ground fault unbalances one phase; a line-to-line fault couples two. The label is a function of which phases share the fault path, and 48 channels record that directly.

CLASS BALANCE



LLL	LL	SLG	LLG
2273	2251	2279	2219

24.60 - 25.26 % BY CONSTRUCTION

DOI 10.5281/ZENODO.21109169
PROTECT-90 / V1.0.0 / CC BY 4.0

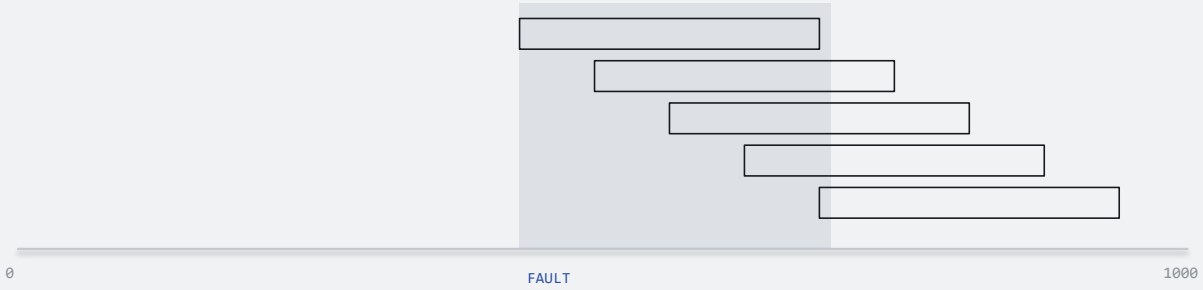
PUBLISHED JULY 2026, FAU ERLANGEN

Source: PROTECT-90 release notes and hv_double_line_90kv_labels.csv. Counts are metadata rows; 828 episodes hold a fault shorter than one 40 ms window.

PREPROCESSING

Split the episode. Then cut the window.

EPISODE 0000 / FAULT 429-695 ms / FIVE WINDOWS SHOWN OF 24



256 SAMPLES (40 ms) / STRIDE 64 / 75% OVERLAP / CAP 24 PER EPISODE

EVERY WINDOW OVERLAPS THE FAULT / MEAN COVERAGE 0.964

01 **SPLIT BY EPISODE**

70 / 15 / 15, stratified on sc_type, seed 42.
6,315 / 1,353 / 1,354 episodes. No episode spans two splits.

02 **CUT WINDOWS INSIDE THE FAULT**

Windows are drawn from the fault interval and clamped to the episode.
An earlier sampler started at t = 0 and reached the fault 7% of the time.

03 **NORMALISE PER CHANNEL**

z = (x - mu) / sigma, fitted on training windows only,
applied unchanged to validation and test.

TWO DIFFERENT UNITS

105,728
TRAINING WINDOWS

6,315
TRAINING EPISODES

One episode yields 2 to 24 windows. Metrics are reported per window; the independence question is on sheet 08.

	EPISODES	WINDOWS
train	6,315	105,728
validation	1,353	22,908
test	1,354	22,519

CORRECTION 001

The first sampler drew windows from t = 0 and reached the fault in 7% of cases; all three models sat at chance. Coverage after the fix: 0.964.

LEAKAGE CHECK: OVERLAP BETWEEN SPLITS = 0 EPISODES (verify_pipeline.py)

FEATURE ENGINEERING

48 channels, four statistics, 192 numbers.

48 × 4 = 192

DESCRIPTORS PER 40 ms WINDOW, COMPUTED IN AMPERES AND VOLTS

RMS	energy in the window	`sqrt(mean(x^2))`
STD	spread about the mean	`sd(x)`
PTP	peak-to-peak excursion	`max(x) - min(x)`
MAXABS	largest excursion, either sign	`max(abs(x))`

DESCRIPTOR FIELD



48 CHANNELS / 4 STATISTICS / 192 CELLS

ONE ROW PER CHANNEL

OBSERVATION 003

Descriptor bars are drawn schematically: they show the layout, not measured values.

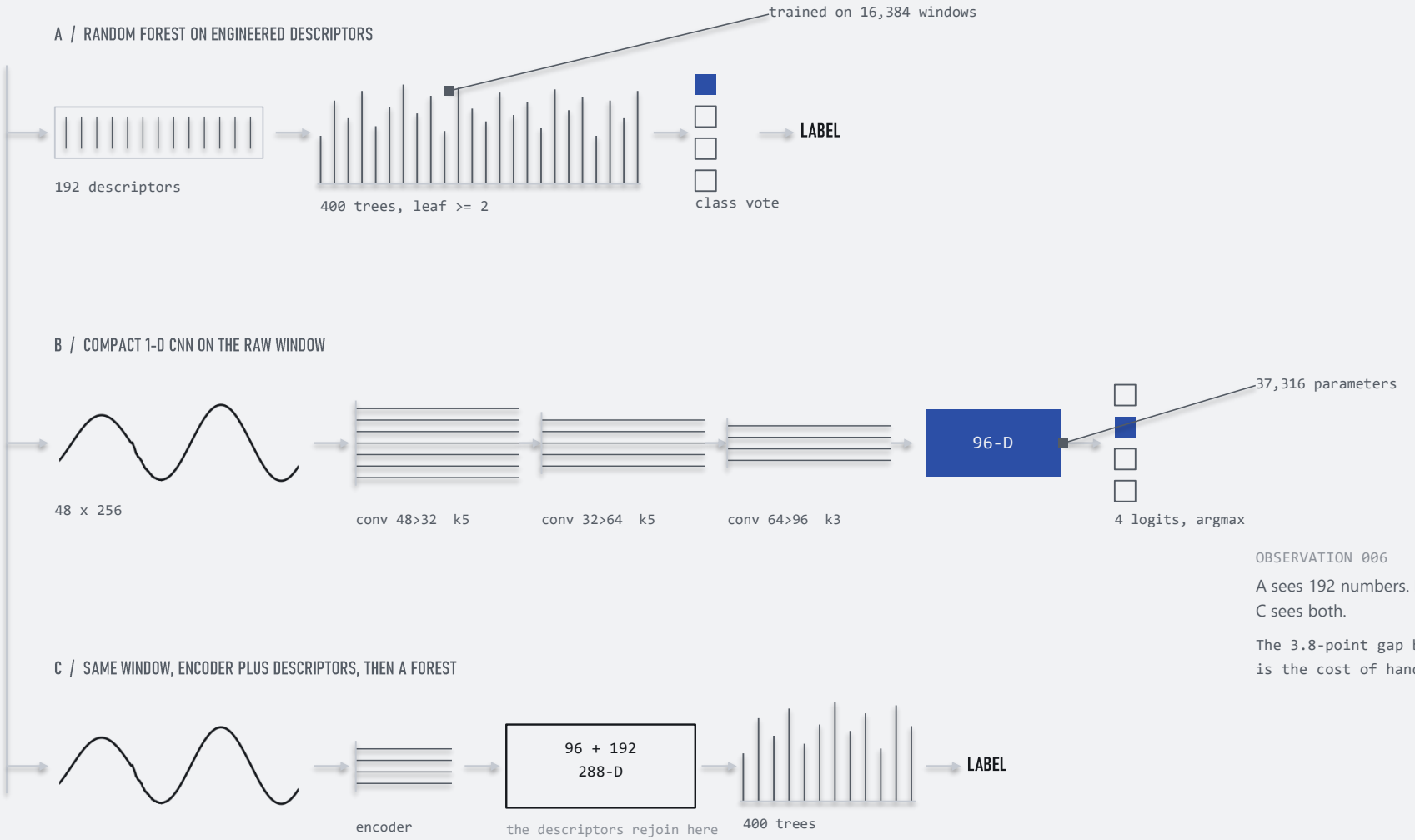
Four numbers per channel discard phase, sequence and waveform shape. That is what the forest gains, and what it pays for.

RF ON DESCRIPTORS 96.11% VALIDATION / 1-D CNN ON RAW WINDOWS 99.92%

Three pipelines, one window.

ONE WINDOW

256 x 48
z-scored



OBSERVATION 006

A sees 192 numbers. B sees 256 samples. C sees both.

The 3.8-point gap between A and B is the cost of hand-built features.

source: hybrid_results/expA,expB,expC_val_metrics.json

VALIDATION SPLIT

22,908 WINDOWS / 4 CLASSES / SEED 42

99.92%

VALIDATION ACCURACY

MACRO F1

0.9992

19 ERRORS / 22,908 WINDOWS



B is the compact 1-D CNN. These windows chose it; the test split played no part in the decision.

VALIDATION / NOT TEST

COMPARISON

MODEL		ACC %	MACRO F1
A	RANDOM FOREST ON ENGINEERED DESCRIPTORS 192-D	96.11	0.9608
B	COMPACT 1-D CNN ON THE RAW WINDOW 48x256	99.92	0.9992
C	CNN EMBEDDING PLUS DESCRIPTORS, THEN A FOREST 288-D	99.89	0.9989

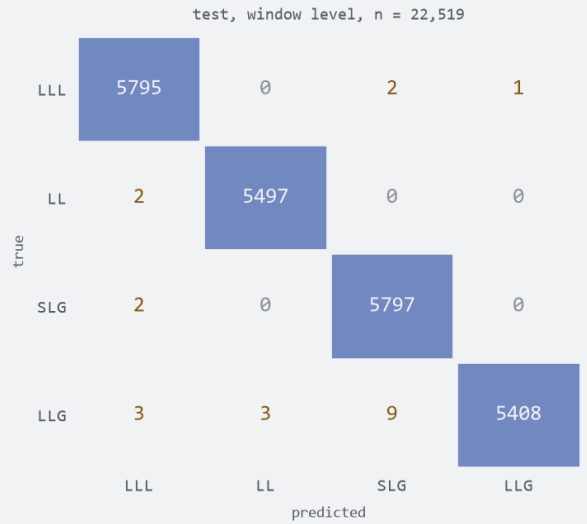
SELECTED: B
MACRO-F1 ON VALIDATION

C trails B by 0.0003 macro-F1. No significance test was run on that difference.

TEST EVALUATION

99.90% of windows. Five of 1,354 episodes.

SELECTED MODEL B / TEST SPLIT / 22,519 WINDOWS / ONE EVALUATION



ROWS / TRUE COLUMNS / PREDICTED

Fifteen of the 22 errors start at LLG. Nine of those land on SLG, so the grounded classes are not cleanly separated from each other.

ACCURACY

0.9990

MACRO P 0.9990 R 0.9990 F1 0.9990

22 WINDOWS OF 22,519 WERE MISPREDICTED

PER CLASS

	P	R	F1
LLL	0.999	0.999	0.999
LL	0.999	1.000	1.000
SLG	0.998	1.000	0.999
LLG	1.000	0.997	0.999

WHERE THE 22 WENT

LLG > SLG	9
LLG > LL	3
LLG > LLL	3
SLG > LLL	2
LL > LLL	2
LLL > SLG	2
LLL > LLG	1

THE OTHER UNIT

99.63 %

EPISODE-LEVEL MACRO F1, 1,354 EPISODES

Each episode is decided by a majority vote of its windows: 1,349 of 1,354 correct, 5 wrong.

MAJORITY VOTE AND MEAN PROBABILITY
PRODUCE THE SAME 1,354 LABELS

Windows inside one episode are not independent. 99.90% is a per-window rate; 99.63% is the rate for a whole record.

REPORTED PER UNIT
NEVER POOLED

SCOPE

What this result is not.

STAGE 1

SIMULATION, COMPLETED

Window classification on PROTECT-90, four classes, episode-wise split, one selected model.

STAGE 2

ROBUSTNESS, NOT STARTED

Noise and SNR sweeps, missing channels, unseen topologies, other sampling rates, calibration.

STAGE 3

FIELD, NOT STARTED

Recorded fault data, CT saturation, evolving faults, inference latency, hardware in the loop.

PROGRESSION OF EVIDENCE



SIMULATION	ROBUSTNESS	FIELD
COMPLETED	NOT STARTED	NOT STARTED

OPEN QUESTIONS, STATED

01 Synthetic data. Near-perfect separability is expected when the fault type determines the phase pattern.

02 A 40 ms window inside the fault is an easy decision. Detection and latency are out of scope.

03 16,384 training windows, one seed, one split. No confidence intervals.

04 A and C were never tested. Only the selected model saw the test split.

FIELD VALIDATION
NOT ESTABLISHED

Nothing in this sheet is a claim about a protection function, a relay setting, or a live network.

THE MODEL IS A RESEARCH CLASSIFIER, NOT A PROTECTION FUNCTION

VIDYUT

What stands, and what it cost.

BUILT	9,022 episodes / 48 channels / 105,728 training windows
EVALUATED	three pipelines / one selected on validation / one test pass
FOUND	99.92% validation, 99.90% test at window level, 99.63% per episode
OPEN	field data, noise, missing channels, latency, other topologies

A 40 ms window drawn from inside a simulated fault is classified almost perfectly from 48 channels. Whether that holds on a feeder with real instrument transformers is a separate experiment, and it has not been run.

[1] Kordowich et al., PROTECT-90, Zenodo, doi:10.5281/zenodo.21109169 [2] Oelhaf et al., arXiv:2606.24298
[3] Breiman, Mach. Learn. 45(1), 2001 [4] Kiranyaz et al., MSSP 151, 2021 [5] Turanli & Bent. Yakut, Appl. Sci. 14(20), 2024